NI4E · PROFESSIONAL KNOWLEDGE SERIES

Engineering Practice & Professional Ethics

Practice Note · 2026

A practical reference for ethical judgement, public safety, transparent engineering decisions and reliable technical records.

1. Professional responsibility

Engineering decisions affect public safety, property, the environment and the reliability of essential services. Professionals should work within their competence, communicate limitations and seek specialist input when a task exceeds their experience.

A defensible decision records the evidence considered, the assumptions made, the options assessed and the reasons for the selected approach.

2. Safety, integrity and accountability

Safety must be considered from design through construction, commissioning, operation and decommissioning. Risk controls should be proportionate to the hazard, documented and reviewed when conditions change.

Professional integrity requires accurate reporting, disclosure of conflicts of interest, protection of confidential information and refusal to falsify test results, measurements, calculations or completion records.

3. Technical documentation

Drawings, calculations, inspection reports, test results, change logs and maintenance records should be traceable to the responsible person, date, revision and applicable approval. Clear records support safe handover, audits, fault investigation and future upgrades.

4. Practical review checklist

Confirm scope, authority and competence. Identify hazards and affected stakeholders. Verify inputs and applicable standards. Record assumptions and independent checks. Communicate residual risk. Keep an auditable record of decisions and approvals.

Practice note

This publication is intended to support professional discussion and continuing learning. Engineers should apply relevant Nigerian legislation, approved standards, employer procedures and project-specific requirements when making technical decisions.